

Variant	Block	Sort	Hash	PROP Time ↓	E2E Step ↓	Global Inst. ↓	Atomic Tx. ↓	Long Scorebd. ↓	Issue Busy ↑
Binning				1.00	1.00	1.00	1.00	1.00	1.00
+ Block	✓			0.95	0.95	0.65	1.00	0.66	1.88
+ Sort	✓	✓		0.94	0.97	0.65	0.97	0.57	1.91
+ Hash	✓	✓	✓	0.90	0.93	0.62	0.93	0.16	3.31

Table C. Block and hash micro-ablation on FlyBrain, normalized to V0 Binning. Lower is better except Issue Busy. RTX 5090; batch=1; 128 steps; event rate=0.01. Atomic Tx denotes global reduction sectors; timing is the CUDA Event median.